

Chen-Yu Chang

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[LinkedIn](#) | [Portfolio](#)

EDUCATION

Boston University | Boston, MA

May 2022

Bachelor of Science in Computer Engineering, Minor: Statistics

GPA: 3.39/4.0

Coursework: Machine Learning, Algorithms, Smart & Connected System, Computer Architecture, Signal & Systems,

High Performance Programming with Multicore and GPUs, Cybersecurity

EXPERIENCE

Product Marketing Engineer | Texas Instruments | Sugar Land, TX

Sept 2022 - Jun 2023, Jan 2024 - Feb 2025

- Led new MCU definition with customer feedback and the need of peripherals and performance on new model vehicles
- Explored and visited worldwide automotive customers and Tier1s to promote and guide future EV architecture
- Expanded market share in new automotive field, including lighting, braking, BMS, zone, and increased 54% revenue
- Released products (F280015x, F28P55x, AM261x) and designed the campaign of application usage and differentiation

Applications Engineer | Texas Instruments | Sugar Land, TX

Jun 2023 - Dec 2023

- Initiated the launch and publications of new product F29x, including datasheet and whitepaper, paper being selected to present in Electronica panel
- Supported customer questions on peripherals, such as DCSM and PWM, within C2000 products on corporate E2E platform
- Conducted and coded temperature testing on F28P65x die to differentiate the defects to push the launch of the product

YouTuber | Remote

Jun 2023 - Present

- Self-learned to film and edit videos sharing tech insights and overseas experience, reaching 3k subscribers with thorough research, script, and plan

Teacher | Advance Foreign Language Institute | Taichung, Taiwan

Jul 2018 - Present

- Taught math, calculus, and physics from age of elementary school to college students with my organization and presentation skills

Video Production Director | FTSANE | Boston, MA (Remote)

Jun 2023 - May 2024

- Directed the team for video promotion on Instagram and YouTube with activity trailers and reviews
- Gained more influence and students in New England area to expand connections for activities and career

RESEARCH PAPERS

The C29 CPU – Unrivaled Real-Time Performance with Optimized Architecture on C2000 MCUs

Nov 2024

- Evaluated key features of new C29x core and rephrase to what customers care
- Compared performance metrics between C28x/C29x cores and competition with a focus on interrupt response time, instruction cycles, and deterministic task execution
- Collaborated with engineers to deliver its pipeline and safety architecture's efficiency improvement
- Suggested industry use cases to showcase the optimization in control performances

Man in the Middle Attack: Principles and Tricks

Dec 2021

- Implemented DoS, ARP spoofing, DNS spoofing, and SSL stripping using Kali Linux, Ettercap, and Wireshark
- Simulated live attacks to capture login credentials on unsecured networks and evaluated weaknesses in HTTP vs HTTPS
- Proposed practical defense strategies including VPN, Dynamic ARP inspection, HSTS preload, and static ARP tables

Graph Traversal Breadth First Search (BFS) Optimization

May 2021

- Developed and benchmarked serial, hybrid, parallel, and GPU-accelerated BFS algorithms, achieving to 144x speedup over baseline implementation
- Enhanced performance through top-down/bottom-up switching heuristics, multithreading, cache blocking, and CUDA-based parallelization

PRESENTATIONS & POSTERS

Ether-Ring with AM26x Simplify Next-Gen Zonal Architecture | CES 2025

Jan. 7 2025

- Proposed a cutting-edge zonal architecture collaborated with Tesla that decrease the weight of wiring by 68% and assembly cost by 13.5%
- Demoed the ether-ring emphasizing the benefit of Ethernet switch integrated in the MCU, which achieve low latency and redundancy safety

- MATLAB & Simulink Model-Based Design for C2000 Real-Time Microcontrollers | TI Webinar** Dec. 11 2024
- Hosted a webinar with MathWorks with 357 audience registered about Model Based Design on MCU support
 - Presented our product portfolio and explained the reference design TIDM-02012 for HVAC application supported with MATLAB
- Reinventing real-time control with advanced CPU architecture | Electronica Europe** Nov. 12 2024
- Showcased the key benchmarks improvements on C29 CPU and how it optimized cost in application systems and development ecosystem

PROJECTS

- Beyond Air Tapping** | Rtangent (Flutter, Unity, Python, Swift, AR) Sep 2021 - May 2022
- Engineered to project the 3D models onto the QR code on the hardware baton, where users can see through iOS app
 - Created an iOS app allowing only authenticated admin to upload contents to project with different QR code
 - Programmed to change QR code when button on the baton is pressed and allow 3D models to be presented in different angle
- Cruise Control** (C, Node.js) Sep 2020 - Dec 2020
- Connected crawler, range sensor ultrasonic sensor, lidar, steering servos, and optical encoder to ESP32
 - Achieved autonomous vehicle maintaining at 0.1~0.4m/s, avoiding obstacles, and completing PID tuning through udp and socket.io data transmission
 - Experimented without failure and exceeded its limit to 1m/s while smoothed out the crawler forward motion
- Smoothie Dispenser** (JavaScript, C/C++, Node.js) Sep 2020 - Dec 2020
- Integrated servos, thermistor, accelerometer, WebCam, Lidar, alphanumeric display to ESP32 to remotely drop and monitor smoothie to a container with a button on their phone
- Dog Cat Classifier** (MATLAB, Machine Learning) Sep 2020 - Dec 2020
- Conducted machine learning, including PCA, perceptron, KNN, to classify 1000 dog and cat image data for each, reaching within 8% error rate
- Greenmate App** (Java) Sep 2019 - Dec 2019
- Designed an Android app that helps ecofriendly activity, including walking instead of driving and recycling
 - Filmed a commercial tutorial and created main character for this app
 - Established score system with built-in pedometer in phone for walking record, and user input of recycling, leading to different character expressions

LEADERSHIP & ACTIVITIES

- Cofounder, Broccoli's in the Fridge** Sept 2020 - Jan 2022
- Fully funded by Sustainability Seed Grant and BU-wide recognized food waste mitigator start-up, Boston, MA
 - Built an iOS app to mitigate food waste by a calendar of best use dates and recipe suggestions by uploading receipt image with Swift and Figma
- Hyper Rocket Team** Oct 2020 - Jan 2021
- Collaborated to design a hyper rocket to reach its maximum height, aiming for 3000m
 - Optimized the weight and structure of the rocket components and electronical system for improving performance through simulations
 - Coordinated the electrical group to design the PCB for self-selected microcontrollers and sensors through Eagle

SKILLS

- Programming Languages: C++, C, python, JavaScript, Swift, Verilog, PHP, asp.NET, CUDA, MATLAB
- Software Proficiencies: VS code, Xcode, Node.js, Flutter, React Native, Eagle, Final Cut Pro, Adobe Illustrator, Unity, Confluence, Tridion
- Others: Sensors (Lidar, accelerometer, thermistor, gyrosensors), WebCam, machine learning, VR/AR